

RudinAnalysis

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Chapter 1

Overview

This blueprint records the current formal development of `RudinAnalysis`. The code follows the chapter order of Rudin's *Principles of Mathematical Analysis*, starting with ordered fields and the construction of the real numbers, then moving to metric topology and the theory of numerical sequences.

The project is intentionally elementary in style: many results are reproved directly from the chosen axioms so that the formal development mirrors the book as closely as possible.

Definition 1. The present formalization is organized into three mathematical blocks: ordered algebraic structures, basic topology of metric spaces, and convergence theory for sequences and subsequences.

Chapter 2

The Real and Complex Number Systems

2.1 Ordered sets and completeness

Definition 2. For a linearly ordered type, we formalize upper bounds, lower bounds, and the corresponding boundedness properties for subsets.

Definition 3. We define supremum and infimum of a set in the usual order-theoretic way.

Definition 4. The least-upper-bound property and greatest-lower-bound property are expressed as predicates on an ordered type.

Theorem 5. *For a linear order, the least-upper-bound property is equivalent to the greatest-lower-bound property.*

Theorem 6. *Any type equipped with the project's completeness structure satisfies the greatest-lower-bound property.*

Theorem 7. *Every nonempty set that is bounded above admits a supremum.*

Theorem 8. *Every nonempty set that is bounded below admits an infimum.*

2.2 Field axioms and ordered fields

Definition 9. The project introduces a custom structure of field axioms, close to Rudin's presentation, in order to rederive elementary algebraic laws from first principles.

Theorem 10. *Within any type satisfying these field axioms, one has additive and multiplicative cancellation laws.*

Definition 11. An ordered field is a linear order together with the project's field axioms and the expected compatibility of order with addition and multiplication by positive elements.

Theorem 12. *In an ordered field, negatives of positive elements are negative, squares of nonzero elements are positive, and inversion reverses order on positive elements.*

2.3 Irrationality and Dedekind reals

Theorem 13. *The formal development proves that there is no rational number whose square is equal to 2.*

Definition 14. The real numbers are constructed as Dedekind cuts of rational numbers.

Theorem 15. *The ordered set of Dedekind reals satisfies the completeness axiom used elsewhere in the project.*

Theorem 16. *Addition on Dedekind reals is commutative and associative, and every element has an additive inverse.*

Theorem 17. *The Dedekind reals carry the expected ordered-field operations: multiplication is commutative and associative, distributes over addition, and every nonzero element admits a multiplicative inverse.*

Chapter 3

Basic Topology

3.1 Metric preliminaries

Definition 18. Open balls, closed balls, and deleted balls are defined directly from the metric.

Definition 19. We formalize bounded subsets, limit points of subsets, and bounded sequences in a metric space.

Theorem 20. *A set in a metric space is open exactly when every point of the set contains a small open ball still lying inside the set.*

Theorem 21. *Every open ball is an open set.*

Theorem 22. *Every deleted ball is open.*

Theorem 23. *If p is a limit point of E , then every open ball around p meets E in an infinite set.*

3.2 Compactness

Definition 24. The project packages the finite-subcover language needed to compare Rudin's compactness with the compactness API from `mathlib`.

Theorem 25. *A subset of a metric space is compact if and only if every open cover admits a finite subcover.*

Theorem 26. *Compactness behaves well under induced topologies, and compact subsets of metric spaces are closed.*

Theorem 27. *For a compact set, any family of closed subsets with the finite intersection property has nonempty total intersection.*

Theorem 28. *Every infinite subset of a compact metric space has a limit point in that compact set.*

Chapter 4

Numerical Sequences and Series

4.1 Convergent sequences

Definition 29. Convergence is defined in a metric space, together with a bundled notion of convergent sequence and its limit.

Theorem 30. *A sequence converges to p if and only if, for every radius, all but finitely many terms lie in the corresponding open ball around p .*

Theorem 31. *Limits of sequences in metric spaces are unique.*

Theorem 32. *Every convergent sequence in a metric space is bounded.*

Theorem 33. *From any limit point of a set one can extract a sequence of points in the set converging to that limit point.*

Theorem 34. *The formalization includes the standard algebra of limits for complex-valued sequences: addition, multiplication, scalar multiplication, and translation by constants preserve convergence.*

4.2 Subsequences and compactness

Definition 35. Subsequences are represented by strictly monotone extraction maps and their induced sequences.

Theorem 36. *Every subsequence of a convergent sequence converges to the same limit.*

Theorem 37. *Compact metric spaces, and in particular bounded sequences in Euclidean spaces, admit convergent subsequences.*

Definition 38. The set of subsequential limits of a sequence is defined as the set of all limits of its subsequences.

Theorem 39. *The set of subsequential limits of a sequence is closed.*

4.3 Cauchy sequences and completeness

Definition 40. The blueprint includes definitions of Cauchy sequences, tail sets, and completeness for metric spaces.

Theorem 41. *A sequence is Cauchy precisely when the diameters of its tails tend to zero; in addition, taking closure does not change diameter.*

Theorem 42. *Under the usual nestedness and diameter hypotheses, a decreasing family of nonempty compact sets has singleton intersection.*

Theorem 43. *Convergent sequences are Cauchy, and Cauchy sequences converge in compact spaces and in Euclidean spaces.*

Theorem 44. *Compact spaces are complete, Euclidean spaces are complete, and closed subsets of complete metric spaces remain complete.*

4.4 Monotone sequences

Definition 45. We define increasing, decreasing, and monotone real sequences.

Theorem 46. *A monotone real sequence converges if and only if it is bounded.*